

Q1 (20 July 2021 Shift 2)

In an electromagnetic wave the electric field vector

and magnetic field vector are given as $\vec{E} = E_0 \hat{i}$

and $\vec{B} = B_0 \hat{k}$ respectively. The direction of propagation of electromagnetic wave is along :

(1) $(\hat{k})$

(2) $\hat{j}$

(3) $(-\hat{k})$

(4) $(-\hat{j})$

Q2 (22 July 2021 Shift 1)

Intensity of sunlight is observed as 0.092 W m^{-2} at a point in free space. What will be the peak value of magnetic field at that point? ($\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$)

(1) $2.77 \times 10^{-8} \text{ T}$

(2) $1.96 \times 10^{-8} \text{ T}$

(3) 8.31 T

(4) 5.88 T

Q3 (25 July 2021 Shift 1)

A linearly polarized electromagnetic wave in vacuum is

$$E = 3.1 \cos[(1.8)z - (5.4 \times 10^6)t] \hat{i} \text{ N/C}$$

is incident normally on a perfectly reflecting wall at $z = a$. Choose the correct option

(1) The wavelength is 5.4 m

(2) The frequency of electromagnetic wave is $54 \times 10^4 \text{ Hz}$

(3) The transmitted wave will be

$$3.1 \cos[(1.8)z - (5.4 \times 10^6)t] \hat{i} \text{ N/C}$$

(4) The reflected wave will be $1 \cos[(1.8)z + (5.4 \times 10^6)t] \hat{i} \text{ N/C}$

Q3 (4)

Hints and Solutions

MathonGo

Q1

Direction of propagation $= \vec{E} \times \vec{B} = \hat{i} \times \hat{k} = -\hat{j}$

Q2

$$I_{\text{avg}} = \frac{B_0^2 C}{2\mu_0} \& \frac{1}{\mu_0} = \epsilon_0 C^2$$

$$I = \frac{B_0^2}{2} \epsilon_0 C^3$$

$$B_0 = \sqrt{\frac{2I}{\epsilon_0 C^3}}$$

$$B_0 = 2.77 \times 10^{-8} \text{ T}$$

Q3

Reflected wave will have direction opposite to incident wave.